
Conestoga College
MATH73235: Advanced Topics in Mathematics
Spring 2025 Midterm 1
June 11, 2025

Exam Duration: 5pm - 7pm.

Student Name (print): _____

Student ID: _____

This exam contains 8 pages (including this cover page) and 6 questions. The total number of possible points is 30. Enter your answers in the space provided.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided.
- **Mysterious or unsupported answers will not receive full credit.** Correct answers should be supported by calculations, explanations, or algebraic work to receive full credit; incorrect answers supported by substantially correct calculations and explanations may still receive partial credit.
- **Provide exact answers** unless otherwise instructed.
- **Simplify all answers as much as possible.** This means that you need to combine like terms, reduce fractions, etc.
- **Be sure to state units for applied problems.**
- **Clearly identify your answer for each problem.**

Question	Points	Score
1	5	
2	5	
3	5	
4	5	
5	5	
6	5	
Total:	30	

Do not write in the table to the right.

First name:

Last name:

Student ID:

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1. (5 points) **Conceptual Questions.** Write an equation, an expression, or a short sentence to describe each of these concepts.

- (a) A vector field $\vec{F}$ in 3-dimensional space is **irrotational**.

Solution.

□

- (b) The main equation from **Green's Theorem** for a vector field $\vec{F} = F_1\mathbf{i} + F_2\mathbf{j}$ that relates a double integral over a region R with a loop integral over the boundary curve C , traversed so that R is on the left.

Solution.

□

- (c) For a curve C parameterized as $\vec{r}(t)$ for $a \leq t \leq b$, the change of variable formula $s(t)$ that reparameterizes the curve using **arclength parameterization**.

Solution.

□

- (d) The **directional derivative** of a scalar field f in the direction of a unit vector $\hat{\mathbf{u}}$.

Solution.

□

- (e) Suppose there is an irrotational vector field $\vec{F}$ in a **domain** D such that over some closed loop inside D , $\oint_C \vec{F} \cdot d\vec{r} \neq 0$. What does this tell us about D ?

Solution.

□

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2. (5 points) Compute a **line integral** of the form $\int \vec{F} \cdot d\vec{r}$, where

$$\vec{F} = [xz, yz, xy], \quad C : \vec{r} = [t, t, e^t], \quad 0 \leq t \leq 5.$$

Hint: Integration by parts formula $\int u \, dv = uv - \int v \, du$.

Solution.

□

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3. (5 points) Find the **curl** of the following vector field $\vec{v}$ at the point $P(3, 4)$:

$$\vec{v} = \frac{1}{\sqrt{x^2 + y^2}} (-y\mathbf{i} + x\mathbf{j}).$$

Solution.

□

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4. (5 points) Suppose $f(x, y, z)$ and $g(x, y, z)$ are two scalar functions. **Prove** the following formula about divergence:

$$\nabla \cdot (f \nabla g) = f \nabla^2 g + \nabla f \cdot \nabla g.$$

Solution.

□

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5. (5 points) Show that the differential form under the integral is **exact** and then find the result of the following line-integral:

$$\int_{(4,0)}^{(6,1)} e^{4y} (2x \, dx + 4x^2 \, dy) .$$

Solution.

□

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6. (5 points) Provide a parameterization $\vec{r}(t)$ of the curve C formed by the intersection of the circular cylinder of radius 1 about the z -axis, and the plane $z = y$. The parametrization must start at $\vec{r}(0) = (1, 0, 0)$ and end at the point $(0, 1, 1)$.

Solution.

□

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Extra Page: Must submit even if unused.